DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, UTTAR PRADESH, LUCKNOW

EVALUATION SCHEME & SYLLABUS
FOR
B. TECH. FOURTH YEAR

FOOD TECHNOLOGY

AS PER

AICTE MODEL CURRICULUM

[Effective from the Session: 2021-22]

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH IV YEAR VII SEMESTER FOOD TECHNOLOGY

SEMESTER- VII **SESSION 2021-22**

Sl. No	Subject Codes	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1	KHU701/KHU702	HSMC -1 #/ HSMC-2	3	0	0	30	20	50		100		150	3
2	KFT 071-073	Departmental Elective-IV	3	0	0	30	20	50		100		150	3
3	KFT 074-076	Departmental Elective-V	3	0	0	30	20	50		100		150	3
4		Open Elective-II [Annexure - B(iv)]	3	0	0	30	20	50		100		150	3
5	KFT 751	FOOD QUALITY EVALUATION LAB	0	0	2				25		25	50	1
6	KFT 752	Mini Project or Internship Assessment*	0	0	2				50			50	1
7	KFT 753	Project I	0	0	8				150			150	4
8		MOOCs (Essential for Hons. Degree)											
		Total	**12**	**0**	**12**							**850**	**18**

*The Mini Project or internship (4 weeks) conducted during summer break after VI semester and will be assessed during VII semester.

DEPARTMENT ELECTIVE – IV

KFT-071 FOOD QUALITY & SAFETY MANAGEMENT
KFT-072 FOOD PROCESSING WASTE MANAGEMENT
KFT-073 NOVEL TECHNIQUES IN FOOD PROCESSING SECTOR

DEPARTMENT ELECTIVE – V

KFT-074 FOOD PACKAGING
KFT-075 SPECIALITY FOODS
KFT-076 FUNDAMENTALS OF FOOD BIOTECHNOLOGY

DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY, LUCKNOW
B.TECH IV YEAR VIII SEMESTER FOOD TECHNOLOGY

SEMESTER- VIII												SESSION 2021-22	
Sl. No	**Subject Codes**	**Subject**	**Periods**			**Evaluation Scheme**				**End Semester**		**Total**	**Credit**
			L	**T**	**P**	**CT**	**TA**	**Total**	**PS**	**TE**	**PE**		
1	KHU801/KHU802	HSMC-2#/HSMC-1#	3	0	0	30	20	50		100		150	3
2		Open Elective-III [Annexure - B(iv)]	3	0	0	30	20	50		100		150	3
3		Open Elective-IV [Annexure - B(iv)]	3	0	0	30	20	50		100		150	3
4	KFT 851	Project	0	0	18				100		300	400	9
5		MOOCs (Essential for Hons. Degree)											
		Total										**850**	**18**

SUBJECT CODE: KFT-071	COURSE TITLE: FOOD QUALITY AND SAFETY MANAGEMENT
EXAM DURATION: 3.0 HOURS	SEMESTER : VII
L: T: P :: 3 : 0 : 0 CREDITS: 3	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide knowledge of Principles of Food Quality & Quality Control**
- **To provide knowledge of Food Safety and its Management**
- **To study the various Food Laws and regulations pertaining to these.**

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the basics of Food Quality, Quality Control, Quality Assurance and Food Safety/
- Understand the basic principles of Sensory Evaluation applicable to foods.
- Understand the basic principles of HACCP, ISO and sanitation in general.
- Understand the laws and regulations pertaining to Food Quality and Safety.

REFERENCE BOOKS / E LEARNING RESOURCES

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Amihud Kramer, Bernard A. Twigg, "Quality Control for the Food Industry: Fundamentals v. 1, The AVI Publishing Company. Inc., Westport, USA	1970[3rd Edition, 2017]
2	J.M.DeMan, P.W.Voisey, V. F. Rasper, D. W. Stanley,":Rheology and Texture in Food Quality", AVI Publishing Company Inc.	1976
3	Pomeranz Y., Meloan, Clifton E.," Food Analysis : Theory and practice", 3rd Edn. IS: 6273 (Part-1& Part II Chapman and Hall).	1994/1995
4	Maynard A. Amerine, Rose Marie Pangborn and Edward B. Roessler, "Principles of Sensory Analysis of Foods", Academic Press, NY	2013
5	https://www.fssai.gov.in › cms › act-2006 [FSSAI Act, 2006]	
6	https://nptel.ac.in/courses/126/105/126105015/ Lecture 2	

7	https://swayam.gov.in/ Category: Engineering & Technology. Sub Category: Agriculture and Food Engineering		
COURSE DETAILS			
UNITS	**S.NO**	**CONTENTS**	**LECTURE HOURS**
I	**1**	Ways of describing of Food Quality. Quality costs. Quality control and Quality Assurance in Food Industry and their functions. Statistical quality control.	**4**
	2	Quality attributes of foods, their classification and significance. Instrumental measurements of important Food Attribute: Appearance, Color, Volume, Density and Specific gravity, Consistency , Defects etc.	**4**
II	**1**	Application of Sensory Evaluation in Quality Management of foods. Objectives Panels involved. Lab Setup and Equipments. Sensory Evaluation Tests. Analysis and Interpretation of sensory scores.	**4**
	2	Rheological and textural characteristics . Texture profile analysis. Correlation between instrumental and Sensory analysis of food quality attributes.	**5**
III	**1**	Principles of food quality assurance, Total Quality Control (TQC) and Total Quality Management (TQM) in Food Industry.	**5**
	2	Food Safety: Concept and Risks Associated with Food: Physical, Chemical, Microbiological and Allergens. General awareness regarding Food safety management, Concept of food traceability for Food Safety,	**5**
IV	**1**	Introduction, prerequisite program of food safety management system, understanding and implementation of Food Safety Management system in food industries, Understanding and Implementation of ISO 22000.	**5**
	2	Global Scenario of Food Safety Management: Introduction to Codex Alimentarius and FSIS, Good Manufacturing Practices (GMP), Good Hygienic Practices (GHP), Good Agricultural Practices (GAP), ISO series, HACCP systems: ISO/ EC 17025. Global standards for contaminants and toxins in foods.	**5**
V	**1**	Food standards and Specifications: Compulsory and voluntary standards. Overview of National and International Regulatory Agencies.	**4**
	2	FSSAI: Definitions, Objectives, Salient Features and Provisions.	**4**
		TOTAL	**45**

SUBJECT CODE: KFT-072	COURSE TITLE: FOOD PROCESSING WASTE MANAGEMENT
EXAM DURATION: 3.0 HOURS	SEMESTER : VII
L: T: P :: 3 : 0 : 0 CREDITS: 3	PRE REQUISITES: NIL

OBJECTIVES:

- **To provide knowledge of Food Processing Industry Waste Management.**
- **To provide knowledge of the various aspects and methods adopted for Waste Management.**
- **To provide knowledge regarding the environmental considerations and Laws pertaining to these.**

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the byproducts associated with various types of food processing industries and its utilization.
- Understand the basic principles of Waste Characteristics pertaining to Food Processing Industry
- Understand the basic principles of Waste Management pertaining to Food Processing Industry
- Understand the laws and regulations pertaining to Waste Management and Disposal.
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REFERENCE BOOKS / E LEARNING RESOURCES

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	The Environment (Protection) Act 1986. (Act No.29 of 1986)	1986, amended 1991
2	AFST(I) & CFTRI Proceedings of the Symposium on By-products From Food Industries: Utilization and Disposal	1980
3	Panoth Abhirami, Lavanya D, Mohan G Naik, Venkatachalapathy N.," Waste to Wealth: Potential of Rice Bran Wax as Edible coating", Indian Food Industry Mag, AFST (India), CFTRI, Mysore, Vol No. 38, Issue No. 1: 38-43	2019
4	J.H. Green, A.Kramer, "Food Processing Waste Management , Westport - Conn : AVI Publishing Company	1979/1981
5	https://swayam.gov.in/ Category: Engineering & Technology. Sub Category: Agriculture and Food Engineering	--
6.	Training Manual Food Safety Supervisor Course Advance (Level 2)	2017

COURSE DETAILS

UNITS	S.NO	CONTENTS	LECTURE HOURS
I	1	Basic considerations: Standards for emission or discharge of environmental pollutants from food processing Industries as per the updated provision of Environment (Protection) Act, 1986.	4
	2	Elements of importance in the efficient management of food processing wastes. Environmental Labs, Government analysts, Sampling Procedures etc as per EPA, 1986.	4
II	1	Characterization and Effective Utilization of by-products from Cereals, Pulses and Oilseeds Industries. Fruits, Vegetables and Plantation products, Fermented foods, Dairy Industry and Milk Products Industry, Fish, Meat, Egg and Poultry processing industries.	8
III	1	Characterization of food Industry effluents, Physical and Chemical parameters, Oxygen demands and their interrelationships, Residues (Solids), Fats, Oils and Grease, Forms of Nitrogen, Sulphur and Phosphorus, Anions and Cations, Surfactants, Colour, Odour, Taste, Toxicity.	6
	2	Unit concept of treatment of Food Industry Effluent. Screening, Sedimentation Floatation as pre - and primary treatments.	4
IV	1	Biological oxidations: Objects, Organisms, Reactions involved, Oxygen requirements, Aeration devices / Equipments used.	5
	2	Systems: Lagoons, Activated sludge process, Oxidation ditches, Rotating biological cont caters and their Variations and advanced modifications. Types, Working Principle, Functioning, Merits and Limitations. Bio Remediation.	5
V	1	Advanced wastewater treatment systems. Physical separations, Micro-strainers, Filters, Ultra filtration and reverse osmosis. Physico-chemical separations: Activated Carbon Adsorption, Ion Exchange, Electro-dialysis and Magnetic Separation.	5
	2	Chemical oxidations and treatment Coagulation and flocculation. Removal of Nitrogen and Phosphorus from Food Processing Waste Waters. Disinfection. Handling and disposal of Sludge. Environmental Considerations.	4
		TOTAL	**45**

SUBJECT CODE: KFT-073	COURSE TITLE: NOVEL TECHNIQUES IN FOOD PROCESSING SECTOR
EXAM DURATION: 3.0 HOURS	SEMESTER : VII
L: T: P :: 3 : 0 : 0 CREDITS: 3	PRE REQUISITES: NIL

OBJECTIVES:

- **To understand about various emerging technologies in food processing**
- **To understand the mechanism of innovative techniques for the improvement of production yield and quality**
- **To understand the applications of novel processing techniques in the processing &preservation of foods.**

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the concepts of High-Pressure Processing and Ultrasound techniques in food processing and microbial deactivation
- Understand the application of microwave and radio frequency wave technology in food processing
- Understand the membrane technology and Super critical fluid extraction process in food sector
- Understand the significance and application of Nanotechnology in food sector

REFERENCE BOOKS / E LEARNING RESOURCES

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Desrosier Norman W," The Technology of Food Preservation", Westport, Conn. AVI Pub. Co. (4th Edition)	1977 / 2004
2	Norman N.Potter, Joseph H.Hotchkiss, " Food Science", 5th Edition, CBS Publishers & Distributors Pvt. Ltd.	2007
3	Sivasankar, B.,"Food Processing and Preservation", Prentice Hall of India Pvt.Ltd., New Delhi.	2002
4	P J Fellows, "Food Processing Technology", 3rd Edition. Woodhead Publishing	2009
5	Gould G. W. "New Methods of Food Preservation", Springer U.S	1995
6	Gustavo V. Barbosa-Canovas,Maria S. Tapia,M. Pilar Cano,,"Novel Food Processing Technologies", CRC press	2004
7	Tewari G, Juneja V.K.,"Advances in Thermal and Non-Thermal Food Preservation", Wiley Blackwell Press	2007
8	https://swayam.gov.in/ Category: Engineering & Technology. Sub Category: Agriculture and Food Engineering	

COURSE DETAILS

UNITS	S.NO	CONTENTS	LECTURE HOURS

I	1	Introduction, scope and application of High Pressure Processing (HPP) of foods. Mechanism of microbial inactivation through HPP. Commercial HPP equipments.	4
	2	Ultrasonics in Food Processing: Principle, properties, equipment and its applications. Microbial inactivation by ultrasound.	4
II	1	Food Processing by Radio Frequency Electric Fields and its applications: Pulse Electric Field (PFE), Moderate Electric Field (MEF).	5
	2	Use of Electromagnetic Radiations in Food Processing: Ohmic heating, IR heating, Microwave heating, Inductive heating.	4
III	1	Advances in Membrane Technology and their industrial applications in Food Processing: Microfiltration, Ultrafiltration, Nanofiltration, Reverse Osmosis. Principle and Equipments involved.	5
	2	Principle, application, advantages and disadvantages of Super Critical Fluid Extraction (SCFE) process. Types of Super critical fluids and their properties, Methods of Extraction. Applications in Food Processing Sector.	4
IV	1	Concept and applications of Nanotechnology in Food Processing, its semblance and contribution in food industry along with advantages and limitations.	4
	2	Significance of Nano particles, Nano composites, Nano emulsions, Nano structured materials and Nano sensors and their applications in Food Processing Sector.	5
V	1	Concept of minimal processing of fruits and vegetables. Natural preservation systems for foods. Synergistic & Antagonistic effects of food mixtures.	3
	2	Novel applications in food analysis: Atomic Absorption Spectroscopy (AAS), Fourier Transform Infrared (FTIR), X-ray Diffraction (XRD), Differential Scanning Calorimetry (DSC), Scanning Electron Microscopy (SEM), e-sensors and biosensors.	7
		TOTAL	**45**

SUBJECT CODE: KFT-074	COURSE TITLE: FOOD PACKAGING
EXAMINATION DURATION: 3 HOURS	SEMESTER: VII
L:T:P: :3: 0: 0 CREDITS: 3	PRE-REQUISITE: NIL

OBJECTIVES:

- To provide basic knowledge of packaging, its types and components, package developments.
- To provide knowledge of different types of packaging materials & their forms used in food packaging
- To provide knowledge of package developments and packaging laws and regulations in food industries
- To provide knowledge about package performance and various testing of packaging materials

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the basic concept of packaging and advancement in food packaging.
- Understand the different types of packaging material and their properties.
- Understand the selection of packages for specific food & agricultural commodities.
- Gain insight of package printing and packaging laws & regulations in food industries.

REFERENCE BOOKS:

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1.	M. Mahadeviah and R.V. Gowramma, " Food Packaging Materials", Tata. Mc Graw Hill co. Ltd. New Delhi	1996/2007
2.	Stanley Sacharow and R.C. Griffin, "Principles of Food Packaging" , AVI Publishing Company	1980
3.	Coles R., McDowell D. and Kirwan, M.J.," Food Packaging Technology", Blackwell Publishing Co.	2003
4.	Proceedings of IFCON-1988	1988

COURSE DETAILS:

UNITS	S.NO	CONTENTS	LECTUREHOURS
I	1	Basic Concepts: Concept of packaging, Functions of a Food Package, Package development factors and Food package development.	4

	2	Newer trends: Aseptic Packaging, Vacuum and gas packaging, Edible packaging, Active and intelligent packaging.	**5**
II	**1**	Cellulosic and Polymeric packaging materials and forms: Food grade polymeric packaging materials, Rigid plastic packages. Films: Multilayer films, Oriented, Co-extruded films , Laminates and Metalized	**5**
	2	Polymeric materials, their mechanical sealing and barrier properties: Cellophane, Olefins, Polyamides, Polyesters, PVC, PVDC, PVA, Inomers, Copolymers, Polycarbonates, Phenoxy, Acrylic and Polyurethane.	**5**
III	**1**	Glass containers: Composition, Properties, Bottle making and Closures for glass containers.	**4**
	2	Metal containers: Bulk containers, Tin-plate containers, Tin free steel containers, Aluminum containers, Latest development in metal cans and protective lacquers.	**5**
IV	**1**	Food product characteristics and package requirement, Selection of materials, Forms, Machinery and methods for fresh produce (Fruits, Vegetables, Egg, Meat and Fish), Edible oils and Fats, Spice and spice products,	**7**
V	**1**	Evaluation of Food Packaging Materials and Package Performance. Evaluation of OTR/ GTR/ WPTR and testing for presence pathogenic microorganisms of concern.	**5**
	2	Package printing, Packaging Laws and Regulations, Food Safety and Environmental issues related to Food Packaging. Disposal of Non Biodegradable Packaging Materials and Environmental Concerns.	**5**
		TOTAL	**45**

SUBJECT CODE: KFT-075	**COURSE TITLE: SPECIALITY FOODS**
EXAM DURATION: 3.0 HOURS	**SEMESTER : VII**
L: T: P :: 3:0:0 CREDITS: 03	**PRE REQUISITES: NIL**

OBJECTIVE:

- To impart the knowledge about the relationship between food, nutrition and health.
- To understand the nutritional requirements and balanced diets for different age groups.
- To understand the concept of modification in normal diets for management and prevention of various diseases.

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the basic concepts of Food Nutrition and Recommended dietary intakes.
- Understand Nutritional requirements for different age groups.
- Understand the concept of Balanced Diets and plan nutritious meals for different age groups.
- Modify the normal diets that can be used for management and prevention of various diseases.
- Understand the beneficial effects of conventional and non-conventional foods and food constituents.

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Benzamin T. Burton, Human Nutrition, Tata Mc Graw Hill	1976
2	P.S. Howe, Basic Nutrition in Health & Disease, W. B. Saunders Co., Philadelphia (7th ed.)	1981
3	Robert E.C. Wildman, Denis M. Medeiros, Advanced Human Nutrition, CRC Press	2014
4	Shubhangini A. Joshi Nutrition and Dietetics, Tata McGraw-Hill Education	2010
5	B. Srilakshmi :Dietetics , New Age International	2014
6	M. Swaminathan, Advanced Text-Book On Food & Nutrition (Vol- I & II), Banglore Printing and Publishing	2012
7	N. Shakuntala Manay and M.Swamy: Food Facts & Principles, New Age International	2008

COURSE DETAILS			
UNITS	**S.NO**	**CONTENTS**	**LECTURE HOURS**
I	**1**	Nutrition, Energy requirements of the body, Recommended Dietary Intakes (RDI) and its uses. Factors affecting nutritional requirement of an individual, Malnutrition.	**4**
	2	Role of nutrients, Effects of deficient or excess intake of the individual essential nutrients.	**4**
II	**1**	Balanced diet and basic food groups; Digestion, absorption and utilization of nutrients in the body.	**4**
	2	Foods for pregnant ladies and nursing mothers, Infant and baby foods, Adolescent / Teen-age foods, Geriatric foods.	**5**
III	**1**	Therapeutic Diets, Modification in texture and consistency of diet, Modification of normal diet during illness and convalescence.	**5**
	2	Diet during energy imbalance and metabolic disorders.	**4**
IV	**1**	Diets recommended and restricted in Gastrointestinal disorders, Cardiovascular diseases, Kidney and Liver disease.	**4**
	2	Dietary modification in Food allergy and Fever & Infections. Diet during Gallbladder and Pancreatic disturbances.	**5**
V	**1**	Beneficial Effects of Spices, Gamma-linolenic acid, Spirulina, Antioxidants and other foods for special medicinal purpose.	**5**
	2	Functional foods & Nutraceuticals, Gluten free foods, Low glycemic index foods, Foods for special dietary use, Organic foods, and Novel & Proprietary foods.	**5**
		TOTAL	**45**

SUBJECT CODE: KFT-076	COURSE TITLE: FUNDAMENTALS OF FOOD BIOTECHNOLOGY
EXAM DURATION: 3.0 HOURS	SEMESTER : VII
L: T: P :: 3 : 0 : 0 CREDITS: 3	PRE REQUISITES: NIL

OBJECTIVES:

- To provide basic knowledge of Food biotechnology and its applications in Food Processing Industry
- To learn about the production of different food products by Microbial means.
- To study the Fundamentals of Tissue Culture and Genetic Engineering

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the importance of Biotechnology in Food processes.
- Understand the microbial production of organic acids, vitamins and bio-pigments.
- Understand the production of fermented food products like beer, wine, tofu and soy sauce.
- Understand the basics of Tissue Culture, Genetic Engineering and Genetically Modified Foods.

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	Crueger and Crueger , Biotechnology: A Textbook of Industrial Microbiology, Sinauer Associates Inc.,U.S.	2017
2	Frazier, W.S. and Weshoff, D.C., Food Microbiology, McGraw Hill Book Co., New York	1988
3	Roger, A., Gordon, B. and John, T, Food Biotechnology, Cambridge University Press	1989
4	Prescott & Dunn Industrial Microbiology, CBS Publishers and Distributors	2004

COURSE DETAILS

UNITS	S.NO	CONTENTS	LECTURE HOURS
	1	Introduction, history, scope and present status of biotechnology in India with reference to Food Technology.	3

I	**2**	Fermentation, types of fermentation: solid and submerged fermentation. Bioreactor, types of bioreactors, Advantages and disadvantages, applications and design aspects of bioreactors used in food industry.	**5**
II	**1**	Biotechnological methods for the production of organic acids, vitamins and bio-pigments (with reference to the microorganisms involved, substrates used, optimum process parameters and their applications)	**5**
	2	Production of single cell proteins, microbial lipids (microorganisms involved, raw materials, advantages, commonly used methods, safety concerns)	**5**
III	**1**	Sources of enzymes, advantages of microbial enzymes, production, extraction and purification of enzymes.	**4**
	2	Biotechnological approach for the production of fermented food products, Beer, Wine, Tofu, Soy Sauce, Tempeh, Kefir.	**5**
IV	**1**	Tissue culture technology and its application, medium used in tissue culture, somatic hybridization, protoplast fusion, secondary metabolite production.	**8**
V	**1**	Structure of DNA and RNA, Mutation, Mutagens, Types of Mutations and applications of Mutations in Strain Improvement.	**5**
	2	Genetic Engineering and applications, Gene Cloning procedures-general outline, Recombinant DNA Technology, Genetically Modified Foods and their applications. Safety issues related to Genetically Modified Foods.	**5**
		TOTAL	**45**

SUBJECT CODE: KFT -751	COURSE TITLE: FOOD QUALITY EVALUATION LAB
EXAM DURATION: 2.0 HOURS	SEMESTER: VII
L: T: P :: 0:0:2 CREDITS: 1	PRE REQUISITES: NIL

OBJECTIVES:

- To provide basic insight into the practical aspects of Food Quality
- To provide knowledge for the assessment of quality of foods from different food categories

COURSE OUTCOME: After successful completion of the course the students will be able to:

- Understand the testing and evaluation of various parameters of food quality
- Understand the different techniques to perform sensory evaluation study of foods
- Understand the concepts of Food quality and its role in food industry

REFERENCE BOOKS

S.NO	NAME OF AUTHORS/BOOKS /PUBLISHERS	YEAR OF PUBLICATION/ REPRINT
1	J.M.DeMan: Rheology and Texture in Food Quality	1976
2	Y.Pomeranz: Food Analysis: Theory and practice	2000
3	M.A. Amerine: Principles of Sensory Analysis of Food	2005 (Reprint)
4	Morris B. Jacobs: The chemical analysis of foods and food products, III Edition, CBS Publishers and distributors New Delhi.	1938(Pub.)2013 (Reprint)
5	S.Ranganna, “Handbook of Analysis and Quality Control for Fruit and Vegetable Products”. S, Tata McGraw-Hill Education	1986

Course details

S.NO	Name of Experiment
1	Sensitivity tests (Threshold/Dilution) to measure Individual Ability for sensory analysis
2	Difference tests to evaluate qualitative and quantitative differences and/or preference between test products
3	Assessment of quality of wheat flour (Water Absorption Power, Gluten Content, Sedimentation Value etc.)
4	Assessment of quality of Bakery Products: Bread, Biscuits, Cakes etc.

5	Assessment of quality of vegetable oils
6	Assessment of quality of Fruit & Vegetable Products: Tomato Products, Jam, Jelly, Marmalades, Squashes& Cordials, Canned Products.
7	Assessment of Quality of Beverages: Tea & Coffee, Carbonated and RTS Beverages.
8	Assessment of Quality of Milk
9	Assessment of Quality of Dairy Products like Butter, Ice-cream, Cheese etc.
10	Assessment of Quality of Fermented Foods like Yoghurt, Sauerkraut etc.